Highlights

Mesh-to-Mesh Transfer Errors in Multistep Unsteady Flow Simulations

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- Global mass correction does not restore a smeared wave profile.
- Shock-width diagnostics must exclude a smeared contact.
- Small transfer defects do not establish conservation during flow evolution.
- Local flux correction preserves mass but permits small relaxed-bound extrema.
- An error in an older BDF2 state changes the next-step lift.

Mesh-to-Mesh Transfer Errors in Multistep Unsteady Flow Simulations

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ARTICLE INFO

Keywords:
Solution transfer
Remeshing
Conservative interpolation
Galerkin projection
Backward differentiation
Finite-volume methods

ABSTRACT

In unsteady simulations with mesh adaptation, solution transfer must preserve spatial structure while supplying the previous time levels required by a multistep integrator. This study compares nearest-neighbor Taylor interpolation, barycentric interpolation, Galerkin projection, and control-volume transfer using static fields, repeated remeshing, and airfoil calculations. The results show why approximation order, boundedness, and conservation cannot be used interchangeably. Under frequent remeshing, first-order nearest-neighbor and linear barycentric interpolation smear the Riemann wave structure, whereas higher-order point interpolation and projection retain it more closely. A global integral correction does not recover lost spatial structure. Conversely, a small cumulative integral defect across transfers does not establish conservation during the intervening flow steps: non-local limiting in Galerkin and control-volume transfer disturbs uniform far-field states and permits mass exchange through open boundaries. A locally conservative flux correction keeps the final mass and energy drift near roundoff and modestly reduces density error for these methods, while allowing small extrema within relaxed bounds and leaving some correction flux unapplied. Controlled restarts with the second-order backward differentiation formula (BDF2) show a separate temporal effect: an error in the older stored state changes the next-step lift even when the latest state matches the reference. In fixed-body Reynolds-averaged Navier–Stokes calculations, smaller transfer-integral defects coexist with load differences that remain sensitive to time-step refinement and iterative convergence. Solution-transfer accuracy therefore depends on the spatial error, its treatment at boundaries, and the history states used in subsequent time integration.

1. Introduction

In unsteady flow simulations with mesh adaptation, remeshing requires the numerical solution to be transferred from the current mesh to the adapted mesh before time integration can resume. Repeated transfers can introduce accumulating errors, so time-accurate adaptation methods must account for solution transfer as well as spatial and temporal discretization [1–3]. A multistep method also requires the stored solutions at previous time levels, referred to here as history states. The question is how errors in these transferred states affect the subsequent solution.

The transfer literature characterizes an operator through three properties of a single field, each of which answers a different question. Interpolation order describes the approximation of smooth fields; boundedness, enforced by limiting, controls new extrema; and conservation constrains selected integrals or moments. Common-refinement and Galerkin approaches provide accurate conservative transfer between nonmatching meshes [4, 5], while constrained formulations preserve additional discrete properties, including norms and weak incompressibility [6]. Conservative remapping instead transfers integrals of a reconstructed source field to individual target cells [7–10]. Conservative Galerkin transfer also supports time-accurate anisotropic adaptation, including high-order implicit formulations with bounded transfer near discontinuities [3, 11]. The three properties are not interchangeable: preserving an integral over the domain leaves the spatial distribution undetermined, and enforcing bounds can alter the raw approximation or projection.

Properties of a single transferred field do not fully characterize the resumption of multistep time integration after remeshing. The time integrator now uses transferred history states rather than states obtained from earlier steps on the target mesh. An error in the older state can therefore perturb the next residual even if the latest state matches the target-mesh reference. Limiting, changes in representation, positivity corrections, and boundary conditions can modify the field further before the solver uses it. Neither an interpolation-error norm for the latest state nor a global conservation defect captures all these effects. Related work on shallow-water equations shows that preserving a discrete equilibrium

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can suppress disturbances introduced by transfer [12, 13]. Here the focus is on history states in compressible-flow calculations and their effect on aerodynamic loads.

The study compares solution-transfer accuracy and its effects on a vertex-centered finite-volume calculation advanced with the second-order backward differentiation formula (BDF2). Four families are considered: nearest-neighbor Taylor interpolation (NN), barycentric interpolation (BAR), continuous Galerkin projection (GP), and conservative control-volume transfer (CV). Numerical suffixes, as in NN1 or GP2, denote nominal order of accuracy for smooth fields. A bounded global correction (GC) isolates the effect of enforcing an integral over the domain. The CV formulation imposes integral constraints over the target median-dual control volumes, allowing comparison with the continuous test functions used by Galerkin projection.

The experiments distinguish approximation error, preservation of spatial profiles and integrals, and the response to transferred history states. Smooth and discontinuous fields test a single transfer; Euler simulations with repeated remeshing reveal cumulative effects. Independent airfoil meshes allow errors in the latest and older BDF2 states to be varied separately. The standard BDF2 coefficients give an exact expression for the residual perturbation, while controlled restarts measure the resulting state and lift changes. These tests connect spatial transfer errors to their consequences for subsequent time integration.

Fixed-body Reynolds-averaged Navier–Stokes (RANS) calculations extend the comparison to turbulence transport and viscous loads. They distinguish differences in the transferred integrals from differences in aerodynamic loads that remain sensitive to numerical settings. All flow experiments use prescribed remeshing on fixed domains, with reference solutions computed independently on the relevant meshes. There are three random mesh-perturbation realizations per amplitude in the Riemann problem, two in each repeated-remeshing airfoil comparison, and one in the RANS sensitivity tests. The Riemann study includes a comparison with locally conservative flux correction. These tests examine solution transfer under controlled remeshing conditions rather than a complete adaptation strategy.

2. Solution-transfer operators

Each method first constructs a field on the target mesh, then applies any limiting, boundary conditions, and admissibility corrections. The methods are nearest-neighbor Taylor interpolation (NN), barycentric interpolation (BAR), continuous Galerkin projection (GP), and conservative control-volume transfer (CV). Numerical suffixes denote nominal order of accuracy, not polynomial degree: BAR2 is second order, while GP2 and CV2 use piecewise-linear fields. The bounded global correction (GC) adjusts the integral of a transferred field and is considered separately.

2.1. Representation and derivative reconstruction

Source and target meshes, A and B, cover the same discretized domain Ω. The nodal state on mesh M is U_M, and its finite-volume mass matrix is $M_M = \mathrm{diag}(w_i^M) \otimes I$, where $w_i^M = |C_i^M|$ is the positive measure of the median-dual control volume C_i^M of vertex i, I is the identity on the state components, and $\otimes$ denotes the Kronecker product. For one scalar component ϕ, the finite-volume integral is

$$Q_M(\phi_M) = \sum_i w_i^M \phi_{M,i}. \tag{1}$$

All methods return values at the flow-solver vertices. Density, momentum density, and total energy per unit volume are transferred as conserved variables; E denotes specific total energy, so the energy variable is ρE. These nodal values are not control-volume averages, even though Eq. (1) is used to calculate their discrete integral.

Higher-order transfer requires derivatives recovered from neighboring values. A weighted least-squares fit uses

$$\phi_j - \phi_i \simeq g_i \cdot r_{ij} + \tfrac{1}{2} r_{ij}^T H_i r_{ij}, \qquad r_{ij} = x_j - x_i. \tag{2}$$

Here g_i and H_i approximate the gradient and Hessian of ϕ at source vertex i, and r_{ij} is the displacement to a neighboring vertex. A degree-one fit omits the Hessian. With smooth data, a full-rank scaled stencil, and controlled mesh geometry, the linear fit gives an $O(h)$ gradient error; the quadratic fit gives $O(h^2)$ gradient and $O(h)$ Hessian errors. Anisotropy, weighting, and stencil conditioning can degrade these estimates [14–17]. Third-order variants use the quadratic reconstruction, and their achieved order is verified numerically.

2.2. Nearest-neighbor and barycentric interpolation

Let $n(j)$ be the source vertex closest to target point x_j^B, and $r_j = x_j^B - x_{n(j)}^A$. Nearest-neighbor Taylor transfer gives

$$(\mathcal{I}_{AB}^{\mathrm{NN1}}\phi_A)_j = \phi_{A,n(j)}, \tag{3}$$

$$(\mathcal{I}_{AB}^{\mathrm{NN2}}\phi_A)_j = \phi_{A,n(j)} + g_{n(j)} \cdot r_j, \tag{4}$$

$$(\mathcal{I}_{AB}^{\mathrm{NN3}}\phi_A)_j = \phi_{A,n(j)} + g_{n(j)} \cdot r_j + \tfrac{1}{2} r_j^T H_{n(j)} r_j. \tag{5}$$

The variants respectively copy the donor value, include its local slope, and include its curvature. For $|r_j| = O(h)$ and sufficiently accurate derivatives, their smooth-field errors are $O(h)$, $O(h^2)$, and $O(h^3)$. The donor changes across source-node Voronoi boundaries, so the transferred value need not vary continuously with target position.

Barycentric interpolation instead uses the source simplex containing the target point. A conforming triangulation or tetrahedralization of the original mesh provides the search geometry, without adding control-volume vertices. In spatial dimension d, a simplex has vertices $x_{i_r}^A$, $r = 1, \ldots, d+1$. Its barycentric coordinates satisfy

$$x_j^B = \sum_{r=1}^{d+1} \lambda_r x_{i_r}^A, \qquad \sum_r \lambda_r = 1, \qquad \lambda_r \geq 0. \tag{6}$$

The same geometric weights define the second-order interpolant and its gradient-corrected extension:

$$(\mathcal{I}_{AB}^{\mathrm{BAR2}}\phi_A)_j = \sum_r \lambda_r \phi_{A,i_r}, \tag{7}$$

$$(\mathcal{I}_{AB}^{\mathrm{BAR3}}\phi_A)_j = \sum_r \lambda_r \left[\phi_{A,i_r} + \tfrac{1}{2} g_{i_r} \cdot (x_j^B - x_{i_r}^A)\right]. \tag{8}$$

BAR2 reproduces affine fields and is bounded by the containing-simplex values. For a quadratic field with exact vertex gradients, the half-gradient term cancels its interpolation remainder; BAR3 is consequently third-order consistent. Shared values and gradients give continuity across conforming simplex interfaces. Neither method imposes a constraint on the integral across transfer.

Derivative limiting changes these properties independently of nominal order. Multiplying the Taylor corrections by $0 \leq \Phi_i \leq 1$ reduces NN2 or NN3 to NN1 when $\Phi_i = 0$, but reduces BAR3 to BAR2. The smooth verification is unlimited; discontinuous tests also use Barth–Jespersen (BJ) limiting [18]. A limiter derived from linear increments does not automatically bound a quadratic reconstruction at every evaluation point. Extrema must therefore be checked for the actual returned field.

2.3. Galerkin projection and cross-mesh integration

Continuous Galerkin projection determines a coupled target field rather than evaluating target vertices independently. Let ϕ_A^h be the piecewise-polynomial source representation and V_B^p the continuous degree-p Lagrange space on the target simplicial mesh. The projected field $\phi_B^h \in V_B^p$ satisfies

$$(\phi_B^h - \phi_A^h, v_B)_{L_2(\Omega)} = 0 \quad \text{for every } v_B \in V_B^p. \tag{9}$$

This is the best L_2 approximation of the source representation ϕ_A^h in V_B^p [4, 5]. Writing $\phi_B^h = \sum_j v_j N_j^B$, where N_j^B are target nodal basis functions and v_j their coefficients, gives

$$H_B v = b, \qquad (H_B)_{ij} = \int_\Omega N_i^B N_j^B \,\mathrm{d}x, \qquad b_i = \int_\Omega N_i^B \phi_A^h \,\mathrm{d}x. \tag{10}$$

The sparse positive-definite matrix H_B couples basis functions sharing target elements. It differs from the diagonal finite-volume matrix M_B used for time integration.

A target element can cross several source elements. Although a target basis function has one polynomial expression there, the source field changes expression at each source-element boundary. The cross-mesh load must therefore be integrated as

$$b_i = \sum_{K_B} \sum_{K_A :\, |K_A \cap K_B| > 0} \int_{K_A \cap K_B} N_i^B \phi_A^h \,\mathrm{d}x. \tag{11}$$

Here K_A and K_B are source and target simplices, and $|K_A \cap K_B|$ is their intersection measure. On each intersection, both expressions are fixed polynomials. These intersections form the common refinement and permit polynomial-exact quadrature: degree two for P^1/P^1 transfer and degree four for P^2/P^2 transfer on affine simplices [19, 20]. Complete, nonoverlapping coverage is required. The intersection geometry and coupled solve distinguish projection from the point searches used by NN and BAR.

GP2 uses P^1 source and target spaces. GP3 constructs a P^2 source representation, including internally reconstructed edge values, and solves for a complete quadratic target field. Only original-vertex values are then supplied to the flow solver; the consequence of this restriction for the finite-volume integral is illustrated in Appendix A. On mixed-element meshes the spaces are P^p on conforming simplicial subdivisions, not Q^p on the original elements. Nominal approximation orders are two and three under smoothness and mesh-stability assumptions; accuracy after vertex restriction is assessed separately.

2.4. Conservative control-volume transfer

Conservative control-volume transfer (CV) constrains the transferred field over the finite-volume cells of the target mesh instead of against continuous basis functions. Let C_i^M be the median-dual control volume of vertex i on mesh M, so that $w_i^M = |C_i^M|$, and let $C_i^M \cap K$ be its piece in element K. Both meshes carry a continuous field $\phi_M^h = \sum_k \phi_{M,k} \psi_k^M$. On triangles and tetrahedra, $\psi_k^M = N_k^M$ is the P^1 hat function. On quadrilaterals, the subdivision joins each edge to the vertex-average center, whose field value is a weighted combination of the original nodal values. Analogous centered subdivisions are used on nonsimplicial three-dimensional elements. The dependent values are chosen to satisfy

$$\int_K \psi_k^M \, \mathrm{d}x = |C_k^M \cap K| \qquad \text{for every element } K \text{ and vertex } k \text{ of } M. \tag{12}$$

For a quadrilateral, the center weight of vertex k is $|C_k^M \cap K|/|K|$. Summation over elements gives $\int_\Omega \psi_k^M \, \mathrm{d}x = w_k^M$, and hence $\int_\Omega \phi_M^h \, \mathrm{d}x = Q_M(\phi_M)$: the field integral equals the finite-volume integral, Eq. (1), on every supported element type.

The source field is thus the continuous interpolant of the nodal solver values rather than a piecewise-constant or piecewise-linear reconstruction on the source control volumes. Its integral equals Q_A, but its integral over a source control volume need not equal the nodal content $w_i^A \phi_{A,i}$. CV therefore preserves the global source integral and the reconstructed field integral over each target control volume, before limiting and boundary treatment. It does not preserve every source control-volume content.

On parallelograms the weights reduce to the arithmetic vertex average, and the representation reproduces affine fields. On quadrilaterals that remain distorted it does not, because Eq. (12) is imposed in preference to affine reproduction.

The n_B target vertex values are defined by

$$\int_{C_j^B} \left(\phi_B^h - \phi_A^h\right) \mathrm{d}x = 0, \qquad j = 1, \ldots, n_B. \tag{13}$$

This is a Petrov–Galerkin condition with trial functions ψ_k^B and the indicator functions of the target control volumes as test functions. Intersection-based P^1-conservative interpolation [9, 10] instead preserves the integral over each target element through an element-wise linear reconstruction, limits its gradient, and recovers vertex values by explicit volume-weighted averaging over the elements incident to each vertex. Equation (13) imposes the integrals over the solver control volumes directly and therefore requires the solution of a sparse system. In matrix form,

$$\widetilde{H}_B \phi_B = \tilde{b}, \qquad (\widetilde{H}_B)_{jk} = \int_{C_j^B} \psi_k^B \, \mathrm{d}x, \qquad \tilde{b}_j = \int_{C_j^B} \phi_A^h \, \mathrm{d}x. \tag{14}$$

Because the functions ψ_k^B sum to one, the row sums of $\widetilde{H}_B$ are w_j^B; by Eq. (12), its column sums are w_k^B. The matrix has nonnegative entries and the element sparsity of H_B. It is symmetric on meshes of triangles and strictly convex quadrilaterals and on tetrahedral meshes: on a simplex, the vertex permutations exchange the median-dual pieces, and on a quadrilateral, the center weight of each vertex is proportional to its median-dual piece. Its entries depend only on target geometry and are evaluated in closed form.

The right-hand side is obtained by integrating the piecewise-linear source field over each target control volume. These integrals are evaluated on intersections of source subelements and target median-dual pieces,

$$\tilde{b}_j = \sum_{K_B \ni j} \sum_{K_A \,:\, |K_A \cap K_B| > 0} \int_{K_A \cap K_B \cap C_j^B} \phi_A^h \, \mathrm{d}x, \tag{15}$$

where K_B ranges over target elements incident to j and K_A over source subelements. Both meshes cover the same polygonal domain in the reported calculations; uncovered measure is below 10^{-20} of the domain. The target coefficients are obtained from Eq. (14) at a relative residual tolerance of 10^{-15}. They are nodal values, not the control-volume averages $\tilde{b}_j / w_j^B$.

Summing Eq. (13) over the target control volumes, which tile Ω, and using the column sums of $\widetilde{H}_B$ gives

$$Q_B(\phi_B) = \mathbf{1}^T \widetilde{H}_B \phi_B = \mathbf{1}^T \tilde{b} = \int_\Omega \phi_A^h \, \mathrm{d}x = Q_A(\phi_A), \tag{16}$$

to within integration and algebraic tolerances. The local statement, Eq. (13), concerns the field: $\int_{C_j^B} \phi_B^h \, \mathrm{d}x$ equals the source content of C_j^B. The nodal content $w_j^B \phi_{B,j}$ represented by M_B differs from it by $(\widetilde{H}_B \phi_B)_j - w_j^B \phi_{B,j}$, which sums to zero over the mesh but does not vanish individually. For $B = A$, $\tilde{b} = \widetilde{H}_A \phi_A$, and the transfer returns ϕ_A. If both representations reproduce affine fields, uniqueness of the solution of Eq. (14) makes the transfer exact for them. CV2 is consequently nominally second order on simplicial meshes and on quadrilaterals that remain parallelograms under refinement; persistent quadrilateral distortion reduces the attainable order. No higher-order CV variant is used.

On simplicial meshes, CV2 and GP2 use the same continuous P^1 trial space and source representation, but different test functions. GP2 uses the continuous hat functions and gives the best L_2 approximation, preserving the global field integral and coordinate moments. CV2 uses control-volume indicator functions and preserves the reconstructed field integral over each target control volume before limiting and boundary treatment. It also preserves the global finite-volume integral, but does not generally preserve coordinate moments or minimize the L_2 error. Both transfers require a coupled solve, so neither has strictly local support. On quadrilateral meshes their representations also differ: GP2 uses a conforming triangulation without added vertices, while CV2 uses the centered subdivision with dependent center values.

The recovery matrix is symmetric positive definite on triangular, strictly convex quadrilateral, and tetrahedral meshes. Its symmetry follows from the median-dual construction; strict diagonal dominance gives positive definiteness on these element types. These properties do not extend generally to mixed three-dimensional cells. The numerical comparisons reported here are two-dimensional, and the principal accuracy limitation is the loss of affine reproduction on persistently distorted quadrilaterals.

CV2 denotes the limited transfer and CV2-u the unlimited construction. The flow comparisons use the same bound-relaxation tolerance as GP2, $\tau = 2 \times 10^{-3}$. Limiting and the subsequent boundary and admissibility treatment change the unlimited field, so the conservation statements must be distinguished from properties of the returned solver state.

2.5. Integral preservation, bounds, and the returned state

When the unconstrained target space contains constants, Eq. (9) preserves the integral of the complete finite-element field. It also preserves consistent coordinate moments represented by the test space. These identities do not imply conservation over arbitrary finite-volume cells, whose indicator functions generally lie outside a continuous finite-element space. CV2 uses these indicators as test functions; Eq. (13) preserves the field integral over each target control volume, and Eq. (16) the finite-volume integral. Neither the Galerkin mass-matrix inverse nor $\widetilde{H}_B^{-1}$ is a local operator: unchanged geometry in a subregion does not guarantee unchanged transferred values there.

For GP2, consistent and finite-volume integrals coincide when $w_i^M = \int_\Omega N_i^M \, \mathrm{d}x$ on both meshes. This condition holds on simplicial meshes; on the conforming triangulation of a quadrilateral, the hat-function integrals generally differ from the median-dual measures. The CV2 representation satisfies the corresponding condition, Eq. (12), on every supported element type. For GP3, source reconstruction and target vertex restriction introduce additional representation differences. An exact quadratic projection can acquire a second-order error in the discrete integral after vertex restriction (Appendix A). Third-order pointwise accuracy therefore does not imply that the defect in the finite-volume integral decreases at third order.

The bounded global correction can enforce Eq. (1) for any base transfer. For a uncorrected target field ϕ^r_B, the correction c satisfies

$$\sum_j w^B_j c_j = Q_A(\phi_A) - Q_B(\phi^r_B), \qquad \ell_j \le \phi^r_{B,j} + c_j \le u_j. \tag{17}$$

The nodal lower and upper bounds are ℓ_j and u_j. The defect is redistributed among nodes with available capacity, with saturated nodes fixed [21, 22]. The requested integral must lie within the weighted bounds. A small integral defect does not bound each correction, and bounds imposed separately on conserved components do not ensure positive pressure.

GP and CV limiting constrain the target field using extrema of the overlapping source vertices. Hard limiting uses these bounds directly; soft limiting widens them by τ times the source range. Clipping is followed by redistribution that retains a selected global integral: the simplicial lumped total for GP and Q_B for CV. Once limiting is active, Galerkin orthogonality and best approximation need not survive. For CV, global redistribution likewise changes the individual target-volume integrals while retaining their sum.

Boundary values are interpolated from the source solution on the matching physical boundary. This replaces the unconstrained transfer there and can alter the local projection or control-volume identities. Where source and target wall vertices coincide, this interpolation retains the source wall condition; a change in the discretized surface can instead introduce a mismatch with the target wall orientation. The reported flow tests retain the boundary vertices, avoiding this geometric contribution.

For GP, CV, and globally corrected transfers, boundary changes and scalar floors are accompanied by integral-preserving redistribution. Remaining inadmissible states are conditioned while retaining the affected patch totals. These operations preserve the selected global integral but generally change local constraints. The distinction matters when interpreting conservation diagnostics: a property of the unlimited formulation cannot automatically be assigned to the state subsequently used by the solver. In addition, globally redistributing a clipping defect can alter otherwise uniform regions, because every vertex with remaining capacity can receive part of the correction. With relaxed bounds, this includes far-field vertices. Transfer conservation alone therefore does not ensure that subsequent open-boundary fluxes retain their undisturbed values.

2.6. Locally conservative flux correction

The Riemann comparisons also use a locally conservative (LC) flux correction, applied to each transfer family in place of its usual limiting, global correction, and boundary-state recovery. Let $\tilde{b}_j$ be the integral of the reconstructed source field over target control volume j, as in Eq. (14). The corrected nodal content is written as

$$w^B_j \phi^{\mathrm{LC}}_j = \tilde{b}_j + \sum_k \alpha_{jk} F_{jk}, \qquad F_{jk} = -F_{kj}, \quad \alpha_{jk} = \alpha_{kj} \in [0, 1]. \tag{18}$$

The fluxes connect vertices sharing a target element and recover the integral-corrected unlimited transfer when fully applied. Limiting the pair fluxes constrains the nodal values while retaining antisymmetry, so their global sum vanishes even when only part of the correction is applied. Local conservation here means that each target nodal content differs from the source integral over that control volume only through exchanges with neighboring volumes; it does not require every such content to remain equal to $\tilde{b}_j$.

The bounded low-order state is $\tilde{b}_j / w^B_j$. Iterative flux-corrected transport [23] adds the admissible part of the correction using bounds relaxed by 2×10^{-3} of the source range. The calculation allows at most 12 limiting passes. Conservation follows from the flux pairing, but the finite pass limit does not guarantee full recovery of the unlimited transfer. This treatment is tested only in the Riemann calculations; the airfoil comparisons use the standard transfer treatment described above.

3. Effect of remeshing on multistep time integration

After remeshing, errors in each transferred history state enter the next multistep equation. Comparing the transferred solution with a reference computed on the target mesh holds the target discretization fixed. The BDF2 coefficients then determine the residual perturbation exactly, while the state and load responses also depend on the spatial operator and the accuracy of the nonlinear solve.

3.1. Reference and transferred history states

Between remeshing events, the semi-discrete equations are

$$M_M \dot{U}_M + F_M(U_M, t) = 0. \tag{19}$$

Here t is physical time and F_M is the assembled spatial residual, including fluxes, source terms, and the imposed boundary treatment. The state includes every transported variable required by the time integrator, including turbulence variables for RANS. Boundary constraints must be imposed consistently on the histories and on the equations advanced in time. At a constant time step, the target BDF2 equation is

$$\Phi_B(Y; V, W) = \frac{M_B}{\Delta t}\left(\tfrac{3}{2}Y - 2V + \tfrac{1}{2}W\right) + F_B(Y, t^{n+1}) = 0. \tag{20}$$

Here V and W are the stored solutions at t^n and t^{n-1}, respectively. The mesh is stationary during each time step. A continuously moving mesh would require compatible geometric histories and a discrete geometric conservation law [24, 25]; Eq. (20) is not intended as that general formulation.

Let U_A^k and U_B^k denote reference solutions computed independently on meshes A and B for the same physical problem. The transferred history states and their differences from the target-mesh reference are

$$\widehat{U}_B^k = \mathcal{I}_{AB}(U_A^k), \qquad \varepsilon^k = \widehat{U}_B^k - U_B^k, \qquad k = n, n-1. \tag{21}$$

The map $\mathcal{I}_{AB}$ includes the corrections needed to make the transferred states admissible. Thus ε^k contains both transfer error and differences between the source- and target-mesh solutions. The reference U_B^k allows both continuations to use the same target discretization; it is not an exact continuum solution. The decomposition in Appendix A.1 makes this distinction explicit.

3.2. Residual perturbation and linearized state response

At a common trial state, the transferred and reference BDF2 equations differ only through their histories, and this difference is exact.

Proposition 1 (Residual perturbation from transferred states). *Let Φ_B be the constant-step BDF2 residual of Eq. (20), and let the transferred histories differ from the target-mesh reference states by ε^n and ε^{n-1}, Eq. (21). Then, for every trial state Y,*

$$\begin{aligned}\Phi_B(Y; U_B^n + \varepsilon^n, U_B^{n-1} + \varepsilon^{n-1}) - \Phi_B(Y; U_B^n, U_B^{n-1}) \\ = -\frac{M_B}{\Delta t}\eta^n, \qquad \eta^n = 2\varepsilon^n - \tfrac{1}{2}\varepsilon^{n-1}.\end{aligned} \tag{22}$$

PROOF. At the common trial state and time, the spatial residual and the term $3M_BY/(2\Delta t)$ cancel. The remaining difference is $(M_B/\Delta t)(-2\varepsilon^n + \varepsilon^{n-1}/2)$. □

The identity follows from the linear dependence of the BDF2 residual on the stored states. It holds for nonlinear F_B and finite discrepancies. Where boundary conditions modify the residual equations, the same modification applies to the predicted perturbation. Predicting the change in the computed solution requires additional assumptions.

The difference between the computed states $\delta^{n+1} = \widehat{U}_B^{n+1} - U_B^{n+1}$ obeys a different relation. Let r and $\widehat{r}$ be the algebraic residuals remaining after the reference and perturbed physical time steps. Subtraction and expansion about the target-mesh reference state give

$$\begin{aligned}A_B\delta^{n+1} &= M_B\eta^n + \Delta t(\widehat{r} - r) - \Delta t\,\mathcal{N}_B(\delta^{n+1}) \\ A_B &= \tfrac{3}{2}M_B + \Delta t J_B, \qquad J_B = \frac{\partial F_B}{\partial U}.\end{aligned} \tag{23}$$

The Jacobian J_B is evaluated at the target-mesh reference solution at t^{n+1}, and $\mathcal{N}_B$ is the nonlinear Taylor remainder. A locally Lipschitz Jacobian gives $\mathcal{N}_B(\delta) = O(\|\delta\|^2)$. If the step is locally invertible, the perturbation is small, and the algebraic errors are negligible, then

$$\delta^{n+1} = R_B\eta^n + O(\Delta t\|\eta^n\|^2), \qquad R_B = A_B^{-1}M_B. \tag{24}$$

This estimate applies locally at the specified mesh, state, time step, and norm. Limiter switching or incomplete iterative convergence can invalidate the linearized state response without invalidating the residual identity, Eq. (22).

A superscript $\star$ denotes a computed steady reference state. Table 1 assigns the same spatial discrepancy $\varepsilon^\star = \mathcal{I}_{AB}U_A^\star - U_B^\star$ to the two stored states independently. Writing the corresponding responses as δ_f, δ_n, and δ_{n-1}, local linearity predicts

Table 1
History perturbations used to separate the BDF2 contributions. The spatial discrepancy $\varepsilon^\star$ is the same within each method; the reference case supplies the target-mesh reference.

Configuration	ε^n	ε^{n-1}	η^n
Reference	0	0	0
Both states	$\varepsilon^\star$	$\varepsilon^\star$	$3\varepsilon^\star/2$
Latest only	$\varepsilon^\star$	0	$2\varepsilon^\star$
Older only	0	$\varepsilon^\star$	$-\varepsilon^\star/2$

$$\delta_n \simeq \tfrac{4}{3}\delta_f, \qquad \delta_{n-1} \simeq -\tfrac{1}{3}\delta_f, \qquad \delta_f \simeq \delta_n + \delta_{n-1}. \tag{25}$$

The older-only case isolates an error that a comparison of the latest states would miss. The imposed discrepancies are scalar multiples of the same field, so these ratios test the linearized response in that direction, rather than the action of R_B on arbitrary fields.

3.3. Aerodynamic response

For a differentiable target-mesh output G_B, define $\Delta G^{n+1} = G_B(\widehat{U}_B^{n+1}) - G_B(U_B^{n+1})$. With g_B^{n+1} the gradient of G_B with respect to the state at the target-mesh reference solution, Eq. (24) yields

$$\Delta G^{n+1} = (g_B^{n+1})^T R_B \eta^n + O(\|\eta^n\|^2). \tag{26}$$

The load response depends on the spatial discrepancy weighted by the target dynamics and output sensitivity, not just its integral. In particular, preserving the source integral imposes, componentwise,

$$\mathbf{1}^T M_B \varepsilon^k = Q_A(U_A^k) - Q_B(U_B^k). \tag{27}$$

The right-hand side need not vanish because the reference states can have different integrals. Preserving the source integral therefore need not bring the transferred solution or its loads closer to the target-mesh reference.

After the first step, both the new state difference and the retained older state affect subsequent evolution. The initial load difference therefore need not bound the later response. Disturbances away from the airfoil can produce delayed responses on acoustic or convective time scales. Galerkin projection and implicit time integration can also introduce small changes outside the perturbed region, so arrival times are defined using finite-amplitude thresholds.

4. Numerical experiments

4.1. Static transfer verification

Static transfer tests exclude flow-solver evolution so that operator properties can be measured directly. The smooth asymmetric field is

$$\phi = \rho_\infty \left[2 + 0.3e^{0.8x/L_x + 0.5y/L_y} \sin(2\pi x/L_x) \sin(2\pi y/L_y)\right]. \tag{28}$$

Here ρ_∞ is the reference density, and L_x and L_y are the coordinate length scales. Structured quadrilateral and independently generated mixed triangular/quadrilateral target meshes have nominal resolutions 64, 128, and 256; source resolution is half the target resolution. The relative nodal density error is $E_2 = [\sum_j (\rho_{B,j} - \rho(x_j^B))^2 / \sum_j \rho(x_j^B)^2]^{1/2}$, with equal vertex weights; it measures point-value accuracy rather than a volume-weighted field error. The effective spacing $h_{\text{eff}} = N_{\text{nodes}}^{-1/2}$ defines refinement ratios on the fixed domain. Pairwise orders are

$$p_{k,k+1} = \frac{\log(E_k/E_{k+1})}{\log(h_k/h_{k+1})}. \tag{29}$$

Here E_k is the relative nodal error and h_k the effective spacing at refinement level k. A least-squares fit of $\log E$ against $\log h$ summarizes each three-level sequence. The discontinuous field has density 1.2 outside and 3.6 inside a circular inclusion of radius $0.25\min(L_x, L_y)$, whose interface crosses the mesh. The discontinuous state has $T = 273(2+\sigma)$ K, $\rho = 1.2(2+\sigma)$, and $(u, v) = (100, 50)\sigma$ m/s, where $\sigma = 1$ inside the inclusion and -1 outside. Momentum and total energy follow from these primitive fields with $R = 287.05$ and $\gamma = 1.4$; the conserved fields are transferred independently. Positive density and internal energy determine admissibility. Unlimited and limited transfers are compared through density overshoot, undershoot, and the source-to-target defect in the discrete integral.

4.2. Repeated remeshing in Euler simulations

The repeated-remeshing test examines whether the behavior of a single transfer predicts cumulative changes in the solution profile. It solves the two-dimensional Euler equations on $[0, 5] \times [0, 1]$. The uniform reference mesh has 501×101 vertices and spacing $h = 0.01$. The gas is ideal, with heat-capacity ratio $\gamma = 1.4$ and unit gas constant. Density, pressure, and temperature are denoted by ρ, p, and T, and Cartesian velocity components by u and v. Initially,

$$(\rho, u, v, p) = \begin{cases} (1, 0, 0, 1), & x < 2, \\ (0.125, 0, 0, 0.1), & x \geq 2, \end{cases} \tag{30}$$

so the left and right temperatures are 1 and 0.8. The upper and lower boundaries are inviscid slip walls. At $x = 0$ and $x = 5$, characteristic boundary conditions prescribe total pressure and temperature $(p_0, T_0) = (1, 1)$ and $(0.1, 0.8)$, respectively, with direction $(1, 0)$ at both ends.

The calculation advances 400 physical time steps with $\Delta t = 10^{-3}$. Interior vertices receive fresh random displacements in x relative to the original mesh; boundaries remain fixed, and geometric distortion does not accumulate between successive meshes. For vertex i, let $a_{i,\min}$ be the smallest altitude of the incident corner triangles. The admissible streamwise displacement is bounded by $a_{i,\min}/s$, where the dimensionless deformation parameter s controls mesh perturbation amplitude. Values $s = 4$ and $s = 2$ are used; $s = 2$ permits twice the maximum displacement of $s = 4$.

Remeshing intervals of ten and two physical time steps compare low- and high-frequency transfer. Both BDF2 history states are transferred at each mesh change. The low-frequency comparison contains 80 transfer evaluations; the high-frequency integral record contains 398. The fixed-mesh numerical reference uses the same initial and boundary conditions; differences from it include transfer error and the change in spatial discretization.

Density profiles are averaged in y using a common sampling procedure. A wide-window diagnostic uses $W_1 = [2.40, 3.00]$, with plateau levels estimated at its ends. The transition position is the 50% crossing, and its apparent width is the distance between the 10% and 90% crossings. Differences Δx and Δw from the fixed-mesh reference are expressed in cells of spacing $h = 0.01$. At high transfer counts, contact smearing enters the left plateau interval $[2.40, 2.46]$, so this apparent width measures combined wave distortion rather than shock thickness. A shock-only diagnostic applies the same crossing definition on $[2.55, 2.85]$, excluding the contact tail. The wide-window tables use 16 sampling lines and 800 probes; the shock-only measure uses the stored eight-line mean profile with the same streamwise resolution. Both are evaluated at step 399 against their corresponding fixed-mesh reference.

Integral changes are measured immediately across each transfer, excluding physical evolution. With $I_\rho = Q_M(\rho_M)$, each increment ΔI_ρ is the target integral minus the source integral. The tabulated cumulative defect is $\sum \Delta I_\rho / I_{\rho,0}$, normalized by the initial mass. It measures transfer alone. Direct final-state drift is instead $[I_\rho(t) - I_{\rho,0}]/I_{\rho,0}$ and includes changes during the CFD steps. For this Sod problem, the exact mass and energy are 2.375 and 5.75: no physical wave reaches the open boundaries by $t = 0.399$. Density error is the median-dual-volume-weighted mean absolute difference from the exact Sod solution [26] at that time.

The LC comparison comprises 36 matched pairs spanning NN1–NN3, BAR2–BAR3, GP2–GP3, and CV2. Every method is tested with intervals of ten and two physical time steps at $s = 4$; NN1, BAR2, GP2, and CV2 also use three realizations at both $s = 4$ and $s = 2$ with the shorter interval. Each pair has identical final meshes and prescribed mesh trajectories. NN1+GC has no LC counterpart. Paired percentage changes in density error are calculated relative to the standard treatment before averaging over cases. Direct mass and energy drift are measured in the final double-precision restart, after the last transfer.

4.3. Euler airfoil tests with controlled history errors

The airfoil calculations test consequences that cannot be isolated in the repeated Riemann problem: the independent contribution of the two BDF2 history states and the evolution of a transfer disturbance into an aerodynamic output.

Euler flow over the National Advisory Committee for Aeronautics (NACA) 0012 airfoil is calculated at freestream Mach number $M_\infty = 0.1$ and angle of attack $\alpha = 6°$, with $p_\infty = \rho_\infty = T_\infty = 1$, $a_\infty = \sqrt{1.4}$, $U_\infty = 0.118322$, and $\Delta t = 10^{-3}$. Here the subscript ∞ denotes freestream quantities, a_∞ is the sound speed, and U_∞ is the speed. The airfoil chord is denoted by c. The source and independent target meshes contain 16,255 and 22,508 vertices, respectively, with 32,077 and 44,583 triangles. Both retain the same 370 airfoil vertices and 63-vertex farfield polygon at $100c$, so the discretized surface is unchanged. Forces are pressure-only, with unit reference chord and area and a moment reference at $x/c = 0.25$; lift and drag coefficients are denoted C_L and C_D. Nonzero steady drag is treated as a numerical load, not skin friction or shock-induced drag.

Each steady reference solution is continued with equal initial BDF2 histories. The history combinations in Table 1 are then formed on the target mesh and compared with its reference continuation at the same physical time step. State differences use the nondimensional volume-weighted norm $\|v\|_{M_B} = (v^T M_B v)^{1/2}$ over the stored mean-flow components. Supplementary tests also report the unweighted Euclidean norm $\|v\|_2$; these norms weight the near-body and large farfield cells differently. Relative superposition defects are

$$e^U_{\mathrm{sup}} = \frac{\|\delta_f - \delta_n - \delta_{n-1}\|_{M_B}}{\|\delta_f\|_{M_B}}, \qquad e^{C_L}_{\mathrm{sup}} = \frac{|\Delta C_{L,f} - \Delta C_{L,n} - \Delta C_{L,n-1}|}{|\Delta C_{L,f}|}. \tag{31}$$

State-norm ratios are nonnegative; force ratios retain their sign.

The solver reports $-\Phi_B$. Initial residuals are compared at identical trial states, before an update. The older-only and reference cases are evaluated at $U^\star_B$; the latest-only and both-states cases are evaluated at $\widehat{U}^\star_B$. The corresponding differences in η are $-\varepsilon^\star/2$ and $+\varepsilon^\star/2$. Double-precision residuals are resolved into wall-normal and tangential components to distinguish the temporal terms from slip-wall enforcement. Additional cancellation and amplitude tests are described in Appendix B.

A repeated-remeshing experiment perturbs vertices at wall distances between $0.01c$ and $20c$, while retaining the boundaries. NN2, GP2, and CV2 use matching deformation realizations, with 39 remeshing events followed by evolution in a 400-step interval. Load differences are formed against the corresponding fixed-mesh trajectory. Slopes per 100 physical time steps describe this finite window; they estimate neither long-time drift nor independent-sample uncertainty.

A separate single-transfer experiment perturbs narrow bands beginning at wall distances $0.05c$, $0.10c$, and $0.20c$. It uses GP2; CV2 repeats the longer $0.20c$-band calculation. Fixed pressure shells measure the root-mean-square (RMS) pressure difference from a common fixed-mesh reference, and force onset is defined by a fraction of the response maximum. For a sampling shell at wall distance r, inside a perturbed band beginning at wall distance d, the low-Mach acoustic scale is $(d - r)/(a_\infty \Delta t) \simeq 845(d - r)$ steps. Immediate disturbances extend up to approximately $0.08c$ inward of the displaced band; these shells are excluded from the threshold-dependent propagation fit.

4.4. Fixed-body RANS and numerical sensitivity assessment

The fixed-body RANS calculations examine whether the differences between transfer methods persist when turbulence transport and viscous loads are included. Repeated remeshing perturbs a nominally steady solution, and the load differences between methods are compared with their sensitivity to numerical settings. The flow conditions are $M_\infty = 0.1$, $\alpha = 6°$, and chord Reynolds number $Re_c = 6 \times 10^6$. The turbulence models are Spalart–Allmaras (SA) and shear-stress-transport (SST) [27, 28]. The coarse mesh has 23,271 vertices, 9,686 quadrilaterals in the boundary layer and wake, 26,851 triangles, and 256 airfoil vertices. The medium and fine meshes contain 32,102 and 48,228 vertices. All three retain the same first off-wall spacing, approximately $5 \times 10^{-6}c$. The farfield is at $100c$, and the quadrilateral region extends to about $0.11c$.

The calculations use the Stanford University Unstructured (SU2) solver [29]. Mean-flow convection uses the Harten–Lax–van Leer contact (HLLC) approximate Riemann solver with a monotonic upstream-centered scheme for conservation laws (MUSCL) reconstruction; turbulence transport uses scalar upwinding without MUSCL reconstruction. The BDF2 equations are solved by dual-time stepping at a Courant–Friedrichs–Lewy (CFL) number of 6 in pseudo-time. The nominal limit is 800 pseudo-time iterations per physical time step. After ten iterations, the stopping test checks $\log_{10} \mathrm{rms}(r_\rho) \le -12$, where r_ρ is the nodal density-equation residual in the solver normalization. The dimensional physical time step $\Delta t_0 = 10^{-3}$ s corresponds to the nondimensional value $\Delta t^*_0 = 0.347224$. A superscript $*$ denotes quantities normalized with the solver reference scales. With $U^*_\infty = 0.1$, the 400-step interval spans approximately 13.9 chord flow-through times, where convective time is measured by tU_∞/c. A first-order startup step is ten times smaller. Each closure has a fixed-mesh reference continued for 800 steps from matching histories.

On the coarse mesh, the steady SST solution reaches the density-residual target, whereas the SA solution reaches its 20,000-iteration limit at -11.2, with lift and drag stationary to 10^{-7} over the final 2,000 iterations. The steady SST solution on the medium mesh stops at approximately -11.1. These residual levels and the reference continuations are distinguished from convergence of the remeshed physical time steps.

The same prescribed mesh sequences are used for NN2, limited GP2, and limited CV2. At each remeshing event, 12,602 vertices may move and 10,669 are fixed. No vertex with solver wall distance below $0.01c$ moves; the geometric mask uses distance to the nearest airfoil vertex, and part of the outer part of the quadrilateral region can move. The nominal RANS deformation uses $s = 4$, and a minimum corner-area ratio of 0.2 preserves valid cells. The boundaries remain fixed. Remeshing is performed every ten physical time steps, independently of the evolving solution.

Both mean-flow and turbulence history states are transferred. SA transports the working variable $\tilde{\nu}$. For SST, k and ω denote turbulent kinetic energy and specific dissipation rate; their density-weighted values are transferred before lower bounds are imposed on the primitive variables. Wall distance and eddy viscosity are recomputed. For a transferred scalar ϕ, the relative integral defect is $[Q_B(\phi_B) - Q_A(\phi_A)]/Q_A(\phi_A)$; all monitored source integrals are nonzero. Integrals of the turbulence variables are not conserved during physical evolution, because their transport equations contain source terms. The SST integral diagnostic is evaluated before the final lower bounds are imposed, so it does not describe the fully corrected state.

The pressure and viscous contributions to total drag are denoted by $C_{D,p}$ and $C_{D,v}$, respectively. For an aerodynamic coefficient C_Q and transfer method m, define the remeshing-induced response and difference between methods by

$$D_{m,Q}(t) = C_{Q,m}(t) - C_{Q,\mathrm{ref}}(t), \qquad P_Q(t) = D_{\mathrm{NN2},Q}(t) - D_{\mathrm{GP2},Q}(t). \tag{32}$$

Pressure and viscous contributions use identical surface quadrature for the remeshed and reference calculations. Over N common physical sample times t_i, the RMS of a scalar history q is $\mathcal{R}(q) = (N^{-1}\sum_i q(t_i)^2)^{1/2}$. For a modified numerical setting c, relative to the nominal setting denoted by subscript 0, sensitivity is measured by

$$S(q;c) = \frac{\mathcal{R}(q_c - q_0)}{\mathcal{R}(q_0)}. \tag{33}$$

This measures the difference between histories, not the fractional change of their RMS amplitudes. The latter is $[\mathcal{R}(q_c) - \mathcal{R}(q_0)]/\mathcal{R}(q_0)$.

The sensitivity assessment uses a common 38-sample window containing three remeshing events. Inner-iteration limits are increased to 1600 for both closures and to 6400 for SST. Time-step refinement uses $\Delta t_0/2$ for both closures and $\Delta t_0/4$ for SA, with identical displacement fields applied at matching physical times and a separate fixed-mesh reference for each step size. The medium and fine steady meshes assess absolute-load sensitivity. A short SA remeshing comparison on the medium mesh uses the same displacement statistics and amplitude limit but an independent nodal deformation field, combining resolution and disturbance changes.

Computational assistance Generative artificial intelligence (AI) tools, ChatGPT (OpenAI) and Claude (Anthropic), assisted the design of verification tests, code development, and numerical post-processing. The tabulated values and data-driven figures are obtained from numerical outputs through the definitions given above, not from generative model predictions. *[Authors: specify the model versions and verify the scope of research assistance and the independent checks performed.]*

5. Results

5.1. Interpolation order verification

The smooth-field tests recover the nominal orders of the NN, BAR, and GP transfers, while CV2 depends on element geometry (Figure 1). NN1 is first order; NN2, BAR2, and GP2 are consistent with second order; NN3 is consistent with third order. BAR3 and GP3 slopes slightly exceed three over this range, approaching approximately 3.2 on the finest pair, consistent with their nominal third-order construction. On the finest mixed grid, the relative errors are approximately 2.6×10^{-7} for BAR3, 3.4×10^{-7} for GP3, and 2.6×10^{-6} for NN3. CV2 is second order on the structured grid, with a fitted slope of 2.03, but reaches only 1.49 on the mixed grid and 1.35 on its finest pair, consistent with the reduced attainable order on quadrilaterals that remain distorted under refinement (Sec. 2.4). It returns the discrete mass to within 4×10^{-15} on both grids, whereas the GP2 defect on the finest grids is 4.3×10^{-12} (structured) and 2.5×10^{-7} (mixed).

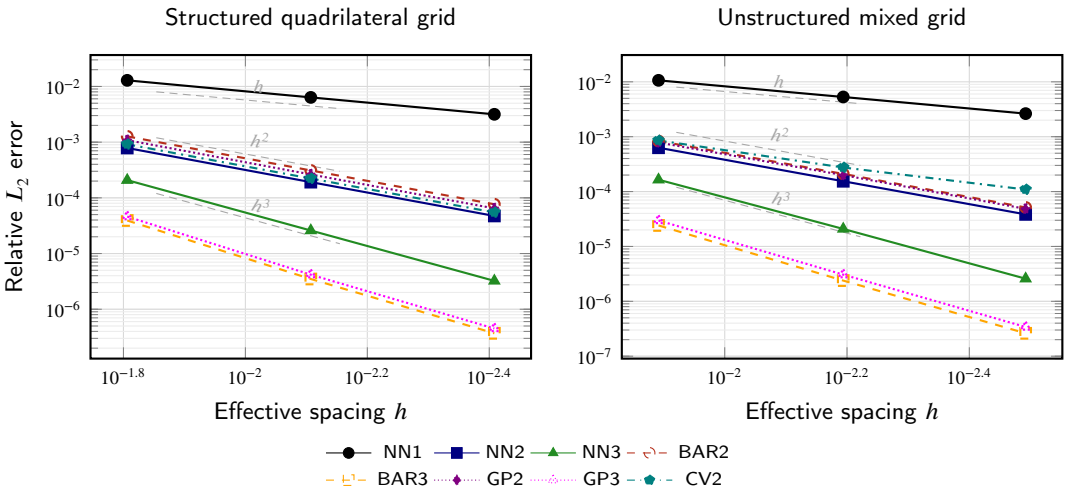


Figure 1: Smooth-field transfer accuracy on structured quadrilateral (left) and unstructured mixed (right) target meshes. The ordinate is the relative nodal density L_2 error; h is the count-based effective spacing. NN, BAR, GP, and CV denote nearest-neighbor Taylor interpolation, barycentric interpolation, Galerkin projection, and control-volume transfer, with numerical suffixes indicating nominal accuracy order. Gray lines are first-, second-, and third-order slope guides, not fitted curves.

5.2. Boundedness and conservation

The circular-inclusion test shows that integral preservation does not ensure a bounded or admissible field (Table 2, structured transfer 64 → 128). Unlimited CV2 preserves the density integral to roundoff while exceeding the source bounds. Unlimited NN2 produces nonpositive internal energy or pressure near the interface and is rejected as inadmissible. These failures concern different properties of the returned state.

Limiting suppresses the measured new extrema, but can change integral behavior. Where Barth–Jespersen limiting removes the NN2 reconstruction, its integral defect rises to the NN1 value. Adding the bounded global correction restores the integral without introducing density extrema. Hard-limited GP2 and CV2 also satisfy both density bounds and near-roundoff integral preservation in this test. Vertex-restricted GP3 remains bounded but retains a nonzero defect in the finite-volume integral, consistent with the representation difference in Appendix A.

Table 2
Representative discontinuous-field transfers. Overshoot and undershoot are excursions beyond the exact density range $[1.2, 3.6]$; the last column is the source-to-target defect in the solver's discrete mass. BJ denotes Barth–Jespersen limiting and GC the bounded global conservation correction. Unlimited NN2 is not listed because its returned state is inadmissible.

Transfer	Overshoot	Undershoot	Integral defect
NN3, unlimited	0.4259	0.2251	-5.52×10^{-7}
NN2, BJ	0	0	-4.46×10^{-3}
NN2, BJ+GC	0	0	2.99×10^{-14}
BAR3, unlimited	0.1766	0.1811	-8.80×10^{-7}
BAR3, BJ+GC	0	0	5.34×10^{-14}
GP2, unlimited	0.1346	0.1357	-2.52×10^{-15}
GP2, hard limited	0	0	1.06×10^{-15}
GP3, hard limited, vertex output	0	0	-4.83×10^{-7}
CV2, unlimited	0.0730	0.1807	5.83×10^{-15}
CV2, hard limited	0	0	6.62×10^{-15}

For a separate smooth transfer, global correction reduces the density-integral defect to order 10^{-15} while changing the density L_1 error by at most 0.8%. The correction changes the accuracy only slightly in this single-transfer test.

5.3. Accumulated broadening under repeated remeshing in Euler simulations

Frequent remeshing smears both the contact and the shock, and a crossing-based width can combine the two effects (Figure 2). After 398 transfers, the smeared contact at $x \simeq 2.370$ enters the wide-window left plateau interval for NN1 and BAR2. Its estimated density can rise from approximately 0.266 to 0.287, moving the 10% crossing into the contact tail. The resulting apparent broadening, exceeding 20 cells in the most diffused cases, therefore includes contact smearing and cannot be interpreted as shock broadening alone. The shock-only diagnostic uses plateaus closer to the shock to separate these effects; Supplementary Section S1 gives the full wide-window comparisons.

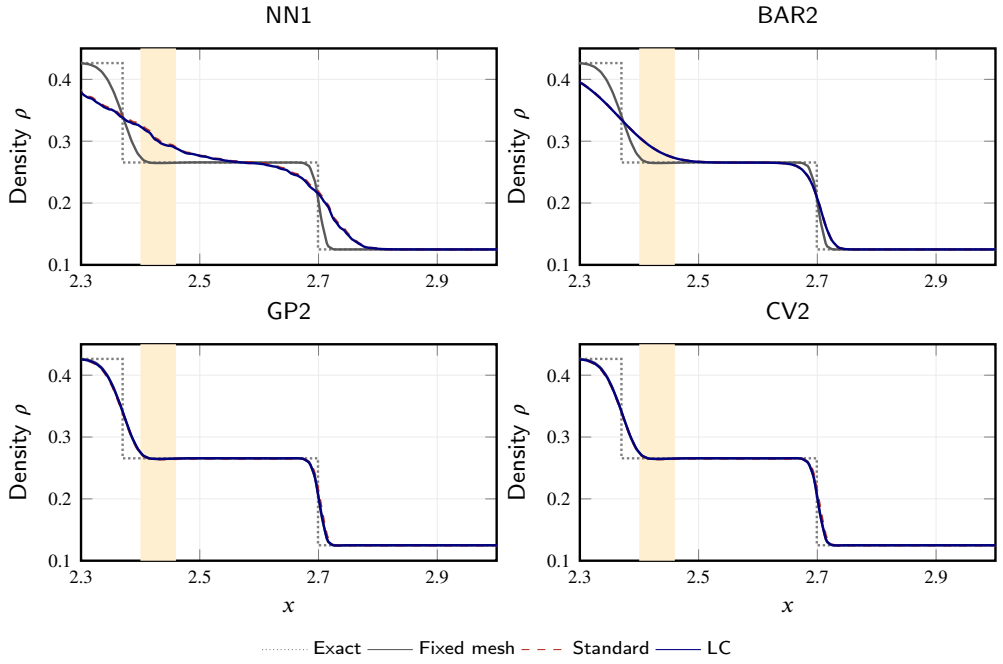


Figure 2: Density profiles after 398 transfers at $t = 0.399$, with remeshing every two steps, $s = 2$, and deformation realization 1. Each numerical curve is an eight-line mean. The exact Sod solution identifies the contact at $x \simeq 2.370$ and shock at $x \simeq 2.699$. Shading marks the left plateau interval $[2.40, 2.46]$ used by the wide-window diagnostic: contact diffusion raises this plateau for NN1 and BAR2. Shock-only widths instead use $[2.55, 2.85]$. Locally conservative flux correction (LC) has little effect on the visible broadening; several standard and LC curves overlap.

Across the three realizations, mean shock-only broadening at 398 transfers is 1.43 cells for NN1 at $s = 4$ and 8.35 at $s = 2$, compared with 1.07 and 2.21 cells for BAR2. The GP2 and CV2 means remain within 0.02 cell of the fixed-mesh shock width (Supplementary Section S5). Their wide-window width changes are also small, at most 0.07 cell in magnitude. These results support better preservation of the wave profile by GP2 and CV2 than by NN1 and BAR2 under frequent remeshing, while distinguishing contact diffusion from shock broadening.

NN1+GC keeps the normalized signed sum of transfer mass changes below 1.0×10^{-13} across all realizations, yet its wide-window profile remains strongly smeared. CV2 also keeps this transfer-only defect near roundoff, whereas GP2 gives values of order 10^{-6}–10^{-5}. These diagnostics distinguish integral preservation during transfer from preservation of spatial structure. They do not establish conservation over the complete simulation, because mass can change during the intervening CFD steps.

5.4. Flux correction, mass balance, and wave accuracy

Direct mass integrals expose a limit of the transfer-only diagnostic (Figure 3a). With the standard treatment, the final relative mass drift of CV2 is approximately 3.0×10^{-7}–4.1×10^{-7} at $s = 4$ and 5.5×10^{-6}–6.4×10^{-6} at $s = 2$, despite its near-roundoff transfer defect. GP2 also has finite final drift, reaching about 2×10^{-6} at $s = 4$. The final-step budgets locate the CV2 change in the CFD step rather than the transfer: for $s = 2$, realization 1, the step-399 mass change is 2.7×10^{-8} of the initial mass, while the subsequent transfer change is negligible. Supplementary Section S5 gives representative budgets.

The GP/CV limiter clips the field and redistributes the integral defect globally in proportion to the remaining capacity within its bounds. With tolerance 2×10^{-3}, nominally constant far-field states can participate. The measured density disturbance reaches 1.4×10^{-4} for $x < 1$, and momentum density at the open boundaries reaches approximately 2×10^{-4}. NN and BAR retain those constant states to roundoff. These measurements connect non-local redistribution, far-field disturbance, and mass exchange through the open boundaries during the CFD steps. They do not isolate the limiter through a limiter-off experiment.

Figure 3b shows how LC affects density error during the run. LC keeps the final relative mass drift below 5×10^{-13} and energy drift below 7×10^{-13} across the 36 matched cases. The mean density-error changes relative to the standard treatment are -3.3% for GP and -2.9% for CV; the corresponding NN and BAR means are $+0.2\%$ and -0.3%. At 398 transfers and $s = 2$, GP2 errors decrease by 7.0–7.9% and CV2 errors by 5.1–5.4% across the three realizations. Their wide-window position errors fall from +0.17–+0.20 cell to 0.00–+0.02 cell. Shock-only broadening changes little (Supplementary Section S5): conservative flux correction improves the mass balance and some solution errors but does not remove diffusion already introduced by the underlying transfer.

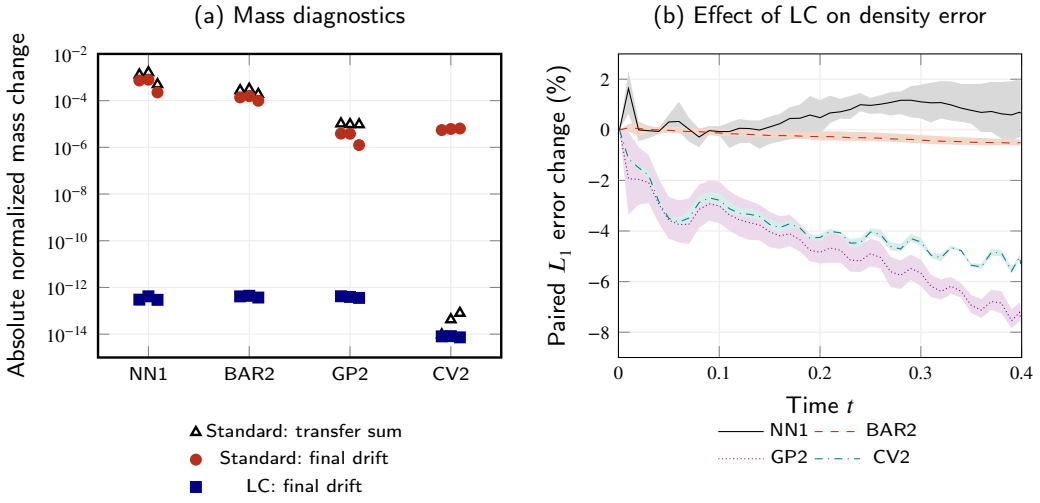


Figure 3: Mass balance and density accuracy for remeshing every two steps with $s = 2$. (a) Each symbol is one of three deformation realizations, offset horizontally for visibility. The absolute signed sum of standard transfer mass changes excludes CFD evolution; final drift uses the direct mass integral after the last transfer. Both are normalized by initial mass. (b) Paired changes $100(E_{1,\mathrm{LC}}/E_{1,0} - 1)$, with density L_1 error measured against the exact Sod solution. Lines are means and shaded bands span the three realizations, not confidence intervals. Samples precede transfer, except the final sample at $t = 0.399$, which follows the last transfer. Negative values indicate lower error with LC.

The relaxed bounds have a cost. LC introduces new density extrema for NN1 and BAR2, which are strictly monotone in the standard Riemann transfers. The extrema remain within the 2×10^{-3} relaxation; BAR2 excursions reach approximately 5.6×10^{-5} of the source range. NN2, NN3, and BAR3 excursions are limited to 2×10^{-3}. The flux correction also reaches its 12-pass limit in 23 of the 36 final transfers, leaving part of the unlimited correction unapplied. The resulting fields remain conservative within the stated tolerance and satisfy the relaxed bounds, but full recovery of the unlimited result is not established. No admissibility repairs are activated in these cases. At slip-wall vertices, LC can introduce wall-normal momentum up to 1.3×10^{-4}, which the flow solver removes in the next step; thus conservation and the bounds do not imply exact satisfaction of the wall condition immediately after transfer.

5.5. Response to errors at the two BDF2 time levels

An error in the older BDF2 state produces a nonzero next-step lift response even when the latest state matches the reference. For every method in Figure 4, perturbing only the older state gives a response opposite in sign to perturbing both states. For GP2, the lift differences are $+2.453 \times 10^{-3}$ and -7.315×10^{-3}, respectively; for CV2, they are $+2.439 \times 10^{-3}$ and -7.272×10^{-3}.

At a common trial state, the measured residual differences agree with the exact BDF2 residual identity to relative errors of approximately 10^{-7} for all tested transfers. Agreement also holds at the 370 slip-wall vertices, where the

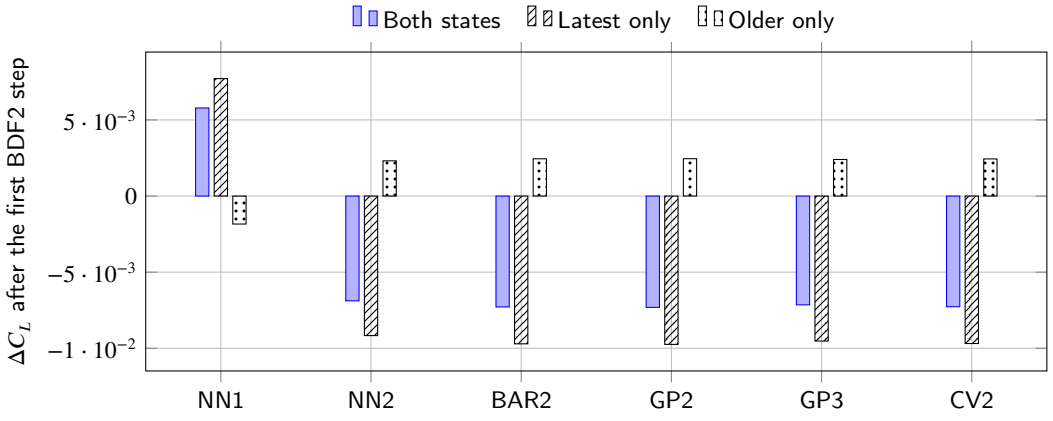


Figure 4: First-step lift-coefficient difference ΔC_L after transfer to the target Euler airfoil mesh, relative to the continuation of the target-mesh reference. NN, BAR, GP, and CV denote nearest-neighbor Taylor interpolation, barycentric interpolation, Galerkin projection, and control-volume transfer; suffixes indicate nominal accuracy order. For each method, the same error field is imposed in both stored states or in one state alone. The opposite signs of the responses with only the older state perturbed and with both states perturbed are consistent with the BDF2 coefficients.

coincident source and target boundaries make the transferred histories compatible with the target slip-wall condition. Supplementary Section S2 reports the separate wall-normal and tangential comparisons.

Global correction barely changes the first-step lift in the NN1, BAR2, and CV2 controlled restarts. For NN1, the response with both states perturbed changes from 5.7873×10^{-3} to 5.7961×10^{-3}, BAR2 from -7.2852×10^{-3} to -7.2849×10^{-3}, and CV2, which already returns the source density integral, from -7.2724×10^{-3} to -7.2765×10^{-3}. These small changes are consistent with the spatial weighting of the load response in Eq. (26).

State differences from the target-mesh reference include transfer errors and differences between the steady solutions computed on the two meshes. The tested second- and third-order methods give norms near 7.0×10^{-4}, with less than 3% spread, whereas NN1 gives 1.47×10^{-3}. This small spread concerns the complete restart error relative to the reference, and does not rank the interpolation errors alone.

Across all eight variants, the state-norm ratio for latest-only to both-states perturbations is 1.3330–1.3333, and the corresponding older-only ratio is 0.3335–0.3341. The relative superposition defects are 1.8–3.5% in the volume-weighted state norm and 0.28–1.49% in lift. These departures from superposition quantify the accuracy of the linearized response for the imposed error field (Eq. (25)). The cancellation and amplitude tests in Appendix B show that the remaining state discrepancy depends on the norm and is not fully explained.

5.6. Accumulated error and delayed loads in Euler simulations

Drag initially grows more slowly with GP2 and CV2 than with NN2 in both Euler realizations. The fitted slopes are 6.31×10^{-6} and 5.89×10^{-6} per 100 physical time steps for GP2, and 5.92×10^{-6} and 5.82×10^{-6} for CV2, compared with 3.95×10^{-5} and 3.84×10^{-5} for NN2. The mean NN2 slope is approximately 6.4 times the GP2 mean and 6.6 times the CV2 mean. The 400-step fitting interval spans only about 0.047 convective and 0.47 acoustic chord times, so these slopes describe initial growth. Lift trends vary more between realizations.

Remeshing in a narrow band away from the airfoil produces a delayed pressure response beyond the region affected immediately by transfer (Figure 5). For the longer calculation with the band beginning at $0.20c$, propagation speeds estimated from 10–75% amplitude thresholds range from 1.09 to 1.27 with GP2 and from 1.10 to 1.30 with CV2. Both ranges include the sound speed $a_\infty = 1.183$; their spread reflects the choice of detection threshold for a single event. In a separate restart test, the NN1 lift difference peaks three steps after the first post-transfer sample at 4.66 times its initial value. The sampled second- and third-order responses peak at the first step and then decay.

5.7. RANS integral defects and load sensitivity

The RANS comparisons first quantify changes in the transferred integrals, then examine the resulting aerodynamic loads. GP2 and CV2 introduce smaller integral defects than NN2 (Table 3). The maximum GP2 density and total-energy defects are roughly two orders of magnitude smaller. The ratios of the separately maximized turbulence-variable defects

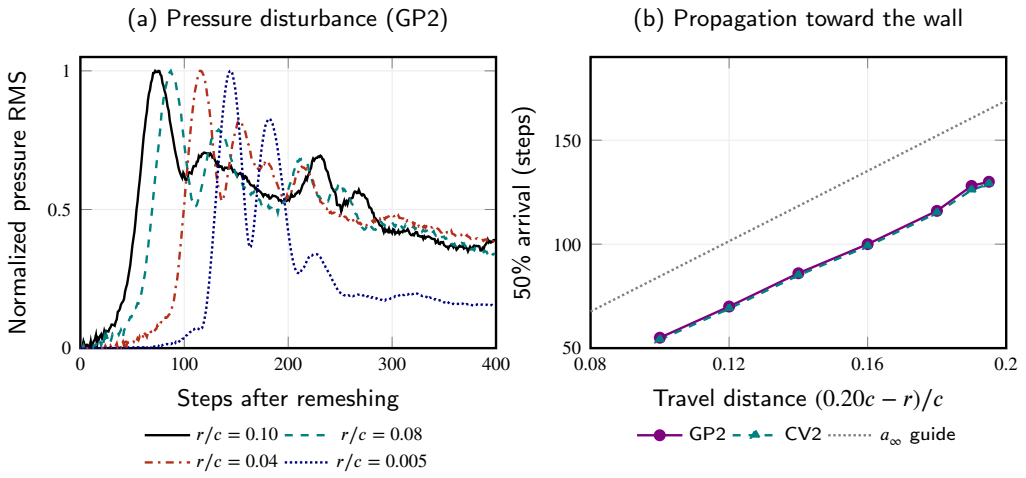


Figure 5: Pressure response to a single Euler remeshing event in the annulus $0.20c \le r \le 0.25c$, where r is distance from the airfoil wall and c is chord. (a) Shell RMS pressure differences from the fixed-mesh reference, normalized separately by each shell's maximum over the 400 post-event samples; shell half-width is $0.004c$. Disturbances reach shells closer to the wall later. (b) First crossings of 50% of each shell's maximum for GP2 and CV2, restricted to travel distances of at least $0.08c$. Connecting lines guide the eye; the dotted line gives travel time at the freestream sound speed. The points describe one event per method, not independent realizations.

are approximately 247 for SA, and 52 and 37 for the SST k and ω integrals. CV2 keeps the mean-flow defects below 1.5×10^{-14} and the turbulence-variable defects below 4.6×10^{-12}. These values are measured immediately across transfer; SST values precede the final lower bounds described in Sec. 4.4.

Table 3
Largest absolute relative integral change across the repeated transfers of the stored RANS states. Each column is maximized separately over the methods' two realizations. SST values are measured before the final lower bounds are imposed. The last column is the NN2/GP2 ratio of the separate maxima.

Closure	Integrand	NN2	GP2	CV2	NN2/GP2
SA	ρ	$8.41 \cdot 10^{-11}$	$4.56 \cdot 10^{-13}$	$7.19 \cdot 10^{-15}$	185
SA	ρu	$3.78 \cdot 10^{-9}$	$5.76 \cdot 10^{-11}$	$9.62 \cdot 10^{-15}$	66
SA	ρv	$1.65 \cdot 10^{-8}$	$8.08 \cdot 10^{-11}$	$1.08 \cdot 10^{-14}$	204
SA	ρE	$9.54 \cdot 10^{-11}$	$5.82 \cdot 10^{-13}$	$1.19 \cdot 10^{-14}$	164
SA	$\rho\tilde{\nu}$	$1.7 \cdot 10^{-6}$	$6.88 \cdot 10^{-9}$	$1.99 \cdot 10^{-12}$	247
SST	ρ	$8.25 \cdot 10^{-11}$	$4.67 \cdot 10^{-13}$	$1.44 \cdot 10^{-14}$	177
SST	ρu	$3.65 \cdot 10^{-9}$	$2.98 \cdot 10^{-11}$	$8.31 \cdot 10^{-15}$	123
SST	ρv	$1.64 \cdot 10^{-8}$	$7.77 \cdot 10^{-11}$	$9.37 \cdot 10^{-15}$	211
SST	ρE	$9.37 \cdot 10^{-11}$	$6.74 \cdot 10^{-13}$	$1.49 \cdot 10^{-14}$	139
SST	ρk	$2.22 \cdot 10^{-6}$	$4.23 \cdot 10^{-8}$	$4.61 \cdot 10^{-12}$	52
SST	$\rho\omega$	$7.98 \cdot 10^{-8}$	$2.15 \cdot 10^{-9}$	$9.29 \cdot 10^{-15}$	37

Over the repeated-remeshing interval, the RMS drag differences from the fixed-mesh references range from 6.09×10^{-5} to 6.79×10^{-5} for the three methods. The RMS differences between NN2 and GP2 are 0.77×10^{-5} to 1.11×10^{-5}; the NN2–CV2 and GP2–CV2 values are 0.66–1.15×10^{-5} and 0.52–0.75×10^{-5}. The ordering changes with turbulence model and realization. The overall responses are therefore similar, while the smaller differences between methods require closer examination of temporal accuracy.

Halving the time step changes the differences between methods substantially (Figure 6). At common physical times, the RMS change in each lift response is 15.1–16.5% of its nominal RMS for SA and 18.8–20.5% for SST. The corresponding drag changes are 8.0–10.2% and 9.6–10.8%. Changes in RMS amplitude alone are much smaller. For the NN2–GP2 difference, however, the changes reach 75.2% in lift and 128.7% in drag for SA, and 146.1% and 110.3%

for SST. The SA quarter-step comparison changes still more. Similar overall amplitudes therefore do not establish convergence of the transient response or a ranking of the methods.

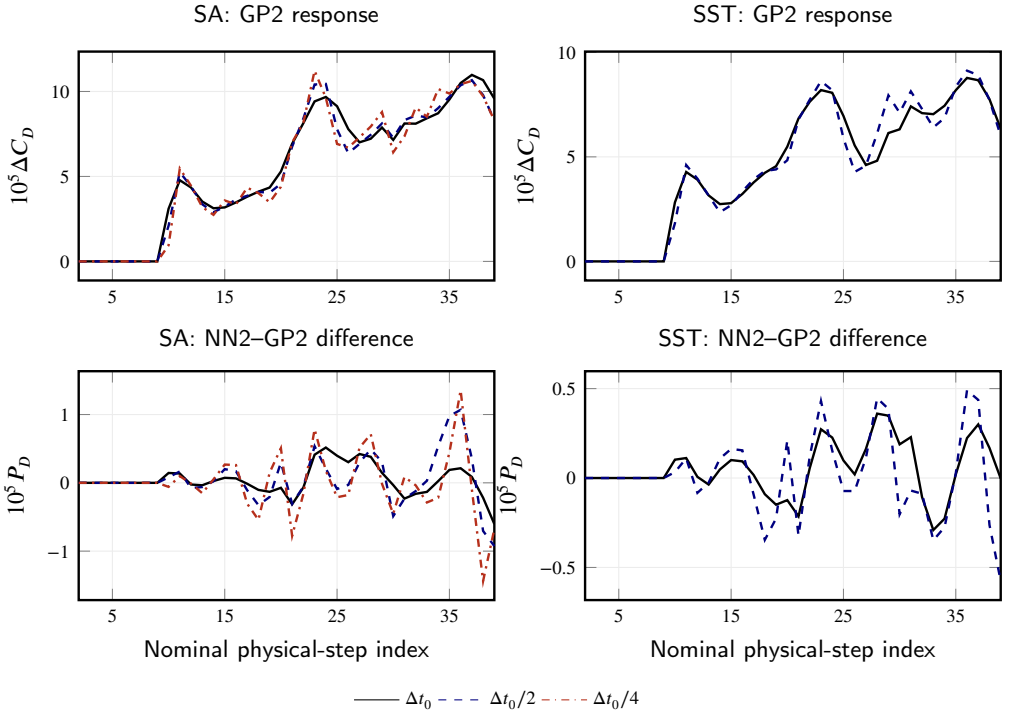


Figure 6: Time-step sensitivity of RANS drag over three remeshing events for deformation realization 1. Columns show the Spalart–Allmaras (SA) and shear-stress-transport (SST) closures. Top: representative GP2 drag differences ΔC_D from the fixed-mesh reference computed at the same time step. Bottom: the smaller NN2–GP2 drag difference, $P_D = C_{D,\mathrm{NN2}} - C_{D,\mathrm{GP2}}$. All curves are sampled at common physical times; the horizontal coordinate is time divided by the nominal step Δt_0. Perturbations are replayed at matching physical times. The smaller difference between methods remains sensitive to the time step; SST also retains inner-iteration error. No SST quarter-step calculation is available.

SA reaches the density-residual target at every physical time step with the reduced time step. Increasing the inner-iteration limit at the nominal step has a smaller, nonzero effect on the difference between methods. For SST, most physical time steps on the perturbed meshes still reach the increased iteration limit, and the $\Delta t_0/2$ calculation is not fully converged. Its time-step sensitivity may therefore include iterative error. Appendix C reports these checks and the more sensitive friction contribution.

The steady RANS calculations show how the absolute loads depend on the mesh, but do not establish an asymptotic refinement sequence or bound the error in the NN2–GP2 difference. The medium-mesh transient also uses a different nodal perturbation, so its response cannot be attributed to refinement alone. Thus GP2 and CV2 have smaller measured integral defects, but the current calculations establish no aerodynamic advantage over NN2. The time-step sensitivity of the NN2–GP2 difference prevents a ranking based on the nominal load responses.

5.8. Spatial profiles and the response after remeshing

The Riemann tests separate three effects that a single conservation or width metric cannot capture. Frequent transfer smears both the contact and the shock; a wide-window crossing can combine the two. Global integral preservation during transfer does not prevent later mass exchange at open boundaries when limiting perturbs the far field. LC suppresses the measured final mass drift and reduces the GP/CV density error, but it does not undo profile diffusion and permits small extrema within its relaxed bounds. The controlled BDF2 restarts isolate a separate temporal effect: errors in either stored state can influence the next solution even when the other state matches the reference.

Differences between independently computed mesh solutions also matter. In the airfoil restarts, the similar errors of the higher-order methods relative to the target-mesh reference include this mesh dependence and do not isolate

interpolation accuracy. Likewise, smaller integral defects in RANS need not imply smaller load errors: global integrals constrain only part of the spatial error, while loads also depend on how that error evolves. This interpretation is consistent with constrained-transfer studies that distinguish integral preservation from other discrete properties [6], and with balance-preserving transfer designed to suppress numerical disturbances [12, 13].

CV2 preserves global integrals during transfer and retains the Riemann profile closely in the tested configurations. Its reduced order on distorted quadrilaterals reflects a trade-off: the representation preserves the finite-volume integral but does not reproduce affine fields on those elements. Limiting and boundary conditions also modify its local integral constraints. The LC comparison demonstrates why these transfer properties must be assessed together with the subsequent flow evolution. It does not establish an LC aerodynamic advantage: the airfoil calculations use the standard treatment, and their RANS load differences remain sensitive to the time step and iterative convergence.

6. Conclusions

Remeshing in an unsteady calculation requires transfer of both the current solution and the history states used by the time integrator. The controlled BDF2 restarts show that an error in the older state changes the next-step lift even when the latest state matches the reference. Its sign is opposite to the response to the same error in the latest state, consistent with the BDF2 coefficients. The residual identity explains these signed contributions; the resulting state and load changes also depend on the spatial operator and iterative convergence.

Integral preservation during transfer, conservation over the full calculation, and preservation of wave structure are distinct. In the Riemann tests, the wide-window 10–90% width includes contact smearing at high transfer counts, while the shock-only measure shows smaller but finite broadening for NN1 and BAR2. GP2 and CV2 retain the wave profile more closely. Their standard non-local limiting can nevertheless perturb uniform far-field states and alter mass during the CFD steps, so a near-roundoff cumulative transfer defect does not demonstrate run-level conservation.

Locally conservative flux correction keeps the final mass and energy drift near roundoff and modestly reduces GP/CV density error, without materially changing shock broadening. Its relaxed bounds allow small extrema, and the finite limiting-pass count can leave correction flux unapplied. Conservation and boundedness therefore constrain the result without guaranteeing recovery of the unlimited transfer or exact wall compatibility.

The fixed-body RANS calculations extend this distinction to turbulence transport and viscous loads. GP2 reduces defects in the turbulence-variable integrals relative to NN2, and CV2 keeps these defects very small at the measured stage. The load differences remain sensitive to time-step refinement, and incomplete iterative convergence adds uncertainty for SST. The slower initial drag growth observed with GP2 and CV2 in the Euler simulations therefore does not establish an advantage in RANS.

The accuracy of the solution after remeshing depends on where transfer errors occur in space and which previous time levels they affect. Conservation is valuable, but its consequences for flow evolution also depend on the discrete representation, the transferred history states, and the numerical accuracy of the subsequent time integration.

Appendix A. Representation and reference-state identities

Conservation of a complete finite-element field need not survive restriction to the solver vertices. For GP3, even an exact quadratic projection is integrated differently when the solver uses only vertex values. To illustrate this, $f(x) = x^2$ on a uniform partition of $[0, 1]$ belongs to the quadratic finite-element space, but integrating only its vertex values with linear nodal weights gives

$$Q_h(f) = \frac{1}{3} + \frac{h^2}{6}. \tag{A.1}$$

This second-order discrepancy comes entirely from vertex restriction; the complete projection remains exact.

Appendix A.1. Discrepancy relative to a target-mesh reference state

A discrepancy from a target-mesh reference solution contains both transfer error and the difference between the mesh discretizations. To distinguish these contributions, let $\Pi_M u^k$ represent a common exact field in the solver representation, with discretization error $e_M^k = U_M^k - \Pi_M u^k$. For a possibly nonlinear transfer map,

$$\varepsilon^k = \underbrace{\mathcal{I}_{AB}(\Pi_A u^k) - \Pi_B u^k}_{\text{transfer consistency}}$$

$$+ \quad \underbrace{\mathcal{I}_{AB}(U_A^k) - \mathcal{I}_{AB}(\Pi_A u^k) - e_B^k}_{\text{source and target discretization contributions}} \quad . \tag{A.2}$$

Only a linear map permits the second line to be written $\mathcal{I}_{AB} e_A^k - e_B^k$. A load difference likewise separates as

$$\begin{aligned} G_B(\widehat{U}_B^k) - G_A(U_A^k) = {} & G_B(U_B^k) - G_A(U_A^k) \\ & + G_B(\widehat{U}_B^k) - G_B(U_B^k). \end{aligned} \tag{A.3}$$

The first difference is dependence of the reference solution on the mesh; the second is the restart response relative to the target-mesh reference. Neither term alone measures the error relative to the exact continuum solution.

Appendix B. Controlled-history diagnostics

Appendix B.1. Signed response to the two histories

The controlled restarts assign the same spatial discrepancy to both stored BDF2 states or to one state alone. Local linearity predicts latest-only and older-only responses of $4/3$ and $-1/3$ times the response with both states perturbed. Table B.1 compares these signed predictions with the computed lift ratios and nonnegative state-norm ratios. The superposition defects quantify the remaining departure from linear response.

Table B.1
Euler BDF2 response ratios and relative superposition defects. Subscripts f, n, and $n-1$ denote perturbations of both states, the latest state only, and the older state only. Ideal directional ratios are $4/3$ and $-1/3$; the state columns show their nonnegative norm ratios.

Transfer	e_{sup}^{U}	$\|\delta_n\|_M / \|\delta_f\|_M$	$\|\delta_{n-1}\|_M / \|\delta_f\|_M$	$e_{\mathrm{sup}}^{C_L}$	$\Delta C_{L,n} / \Delta C_{L,f}$	$\Delta C_{L,n-1} / \Delta C_{L,f}$
NN1	$1.8 \cdot 10^{-2}$	1.3333	0.3335	$1.5 \cdot 10^{-2}$	1.3329	−0.3179
NN2	$3.5 \cdot 10^{-2}$	1.3331	0.3341	$4.2 \cdot 10^{-3}$	1.3318	−0.336
NN3	$3.4 \cdot 10^{-2}$	1.3331	0.334	$5.3 \cdot 10^{-3}$	1.3309	−0.3362
BAR2	$3.4 \cdot 10^{-2}$	1.3332	0.334	$3.1 \cdot 10^{-3}$	1.3328	−0.3358
BAR3	$3.3 \cdot 10^{-2}$	1.3332	0.334	$2.8 \cdot 10^{-3}$	1.3329	−0.3357
GP2	$3.4 \cdot 10^{-2}$	1.333	0.334	$3 \cdot 10^{-3}$	1.3323	−0.3353
GP3	$3.4 \cdot 10^{-2}$	1.3331	0.334	$3.1 \cdot 10^{-3}$	1.3327	−0.3358
CV2	$3.4 \cdot 10^{-2}$	1.333	0.334	$3.1 \cdot 10^{-3}$	1.3323	−0.3354

The residual comparison at a common trial state also includes the slip-wall equations. Because source and target wall vertices coincide, the transferred histories retain the source wall state and the normal residual perturbation is negligible. Supplementary Section S2 gives the component-wise residual comparisons.

Appendix B.2. Cancellation of the initial residual perturbation

For an imposed spatial discrepancy e, setting $\epsilon^n = e/4$ and $\epsilon^{n-1} = e$ cancels the initial residual perturbation, $\eta^n = 0$. Under the same boundary treatment, the reference and perturbed BDF2 equations are then identical. An exact solve on the same local solution branch would return the same next state.

The computed lift differences are of order 10^{-9}, but the volume-weighted state differences remain 4.3–9.0% of the imposed discrepancy (Table B.2). Small load differences therefore do not establish state agreement. The iterative error has not been independently quantified, so this experiment does not verify cancellation of the computed state response. The exact residual identity and its separate common-trial test are unaffected by this limitation.

The initial residual probes in this cancellation experiment use different trial states and consequently include a difference in the spatial residual. Their interpretation is given in Supplementary Section S2. Cancellation also concerns only the first step: even if its state difference vanishes, the retained older history can force subsequent evolution.

Appendix B.3. Limits of the directional approximation

Reducing the imposed discrepancy strengthens superposition in lift and in the Euclidean state norm for BAR2, GP2, and CV2. The volume-weighted norm behaves differently: its absolute remainder decreases slowly while the relative remainder increases from about 3.4% to 8.7–9.2%. Thus the observed coefficient ratios support a directional

Table B.2
Cancellation of the initial residual perturbation. Both history states are perturbed but $2\varepsilon^n - \varepsilon^{n-1}/2 = 0$. First-step force differences and state differences use the same reference continuation; both state norms are shown to avoid interpreting a small load difference as complete state agreement.

Transfer	ΔC_L	ΔC_D	$\|\delta\|_2/\|e\|_2$	$\|\delta\|_M/\|e\|_M$
CV2	$7.59 \cdot 10^{-9}$	$-5.54 \cdot 10^{-9}$	$2.43 \cdot 10^{-4}$	$8.98 \cdot 10^{-2}$
GP2	$7.64 \cdot 10^{-9}$	$-5.54 \cdot 10^{-9}$	$2.43 \cdot 10^{-4}$	$8.98 \cdot 10^{-2}$
NN1	$2.11 \cdot 10^{-9}$	$-4.1 \cdot 10^{-9}$	$2.1 \cdot 10^{-4}$	$4.34 \cdot 10^{-2}$

approximation, but not a uniform quadratic remainder in every norm. Supplementary Section S3 contains the full amplitude comparison.

Appendix C. RANS numerical sensitivity

Appendix C.1. Inner iterations and time-step refinement

Table C.1 separates achieved residual criteria from iteration limits. Rows for the nominal setting cover the full time histories with repeated remeshing; other rows cover the short sensitivity calculations. SA reaches the density criterion throughout the 1600-iteration sensitivity window and at every half- and quarter-step point. SST reaches the iteration limit on 28–30 of 38 steps even with 6400 inner iterations, and on 56–60 of 78 half-step points. These SST references remain not fully converged.

Table C.1
Iterative convergence for the first mesh-perturbation realization. $N_{\max}$ is the iteration limit. Separate columns give the number of steps reaching that limit for each method. Nominal rows cover the full interval; other rows cover the short sensitivity window. Supplementary Section S4 gives the median final density residuals for each method.

				Steps at limit		
Closure	$N_{\max}$	$\Delta t/\Delta t_0$	Steps	GP2	NN2	CV2
SA	800	1	398	152	155	147
SA	1600	1	38	0	0	0
SA	800	1/2	78	0	0	0
SA	800	1/4	158	0	0	0
SST	800	1	398	333	324	336
SST	1600	1	38	28	28	30
SST	6400	1	38	28	30	29
SST	800	1/2	78	56	58	60

Increasing the iteration limit changes the SA lift and total-drag time histories by at most 0.3% of their own RMS and their smaller differences between methods by 8.4–10.9% (Table C.2, Eq. (33)). For SST, the comparison with an iteration limit of 6400 changes the lift difference between methods by 67.7% of its RMS at the nominal setting. Residual stagnation accompanies this sensitivity without identifying a unique cause in the turbulence equations.

Table C.2
Effect of increasing the inner-iteration limit, relative to the nominal setting. Values are $100S$ from Eq. (33); individual columns give the largest NN2, GP2, or CV2 sensitivity, and difference columns refer to the NN2–GP2 difference. All entries are percentages.

		Individual responses		NN2–GP2 difference	
Closure	Iteration limit	C_L	C_D	C_L	C_D
SA	1600	0.3	0.2	10.9	8.4
SST	1600	4.5	1.4	13.7	6.0
SST	6400	8.5	2.6	67.7	25.5

In the SA step-refinement sequence, the RMS of the lift difference between methods is approximately 1.21×10^{-5}, 1.60×10^{-5}, and 2.08×10^{-5}; the drag values are 2.19×10^{-6}, 3.77×10^{-6}, and 4.55×10^{-6}. Although individual RMS amplitudes remain closer across the same samples, the time histories of the differences between methods are unresolved. Friction drag is more sensitive than total drag: the SA quarter-step history differences are approximately 76–92% of the RMS friction-drag response at the nominal setting. The smaller total-drag sensitivity consequently leaves convergence of the separate drag components unestablished.

Appendix C.2. Steady mesh sensitivity

The steady meshes in Table C.3 preserve the first off-wall spacing while changing the remaining resolution. Lift, total drag, and pressure drag decrease monotonically; friction drag varies nonmonotonically. For this anisotropic family, the spacing estimated from the vertex count describes mesh density without establishing a common asymptotic error expansion. The comparison reports absolute-load sensitivity, with no extrapolated loads or discretization-uncertainty estimate; it does not quantify the error in the load difference between transfer methods on a common mesh.

Table C.3
Steady aerodynamic loads without remeshing on the coarse, medium, and fine viscous meshes. The vertex-count estimate $h/h_c = \sqrt{N_c/N}$ uses the coarse-mesh vertex count N_c as reference; it is not a refinement ratio common to all directions. The same first off-wall spacing is used on all levels.

Closure	Mesh	Vertices	h/h_c	C_L	C_D	$C_{D,p}$	$C_{D,v}$
SA	Coarse	23,271	1	0.67626	0.010229	0.003559	0.006671
SA	Medium	32,102	0.851	0.67084	0.010048	0.003401	0.006647
SA	Fine	48,228	0.695	0.66854	0.009996	0.00333	0.006665
SST	Coarse	23,271	1	0.67561	0.009858	0.003469	0.006389
SST	Medium	32,102	0.851	0.6686	0.009705	0.003321	0.006385
SST	Fine	48,228	0.695	0.66639	0.009644	0.003247	0.006397

Appendix C.3. Remeshing response on a different mesh

A separate comparison changes both mesh resolution and the imposed disturbance. On the medium mesh, the short SA remeshing interval gives RMS lift responses for individual methods near 1.07×10^{-4} and drag responses near 5.39×10^{-5}; the differences between methods are 8.77×10^{-6} and 9.94×10^{-7}. The methods remain relatively close, but the new node positions also change the random displacement field. Resolution alone cannot explain the smaller response. Five to seven of the 38 physical time steps reach the nominal inner-iteration limit.

Data availability

Supplementary material contains the full Euler comparisons and extended history and convergence diagnostics. The numerical tables supporting the figures and comparisons, together with the post-processing routines needed to reproduce the reported reductions, are available in the public data repository. The archive does not include every flow-field output from the simulations.

Funding

[Authors: supply funders and grant identifiers, or confirm that there was no specific funding.]

Contributor Roles Taxonomy (CRediT) authorship contribution statement

[Authors: assign CRediT roles to each named author.]

Declaration of competing interest

[Authors: supply the declaration of financial and non-financial competing interests.]

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

ChatGPT (OpenAI) and Claude (Anthropic) were used to assist literature synthesis, organization, scientific editing, and the preparation of the manuscript and data-driven figures. *[Authors: confirm the tools and versions used, describe human review and verification, and provide the final statement accepting responsibility for the published content.]*

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